

Assignment 3 – Optimization of a City Transportation Network (Minimum Spanning Tree)

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1. Introduction

The purpose of this assignment is to optimize a city's transportation network using two Minimum Spanning Tree (MST) algorithms – Prim’s and Kruskal’s. The objective is to minimize the total road construction cost while ensuring that all districts are connected. This report summarizes the input data, the results of both algorithms, their efficiency comparison, and conclusions.

2. Input Data Summary

The input data was provided in the JSON file `ass\_3\_input.json` and contains two graphs representing city districts and potential roads.

Graph ID	Vertices	Edges	Example Nodes	Example Edges
1	5	7	A, B, C, D, E	(A–B, 4), (B–C, 2), (D–E, 6)
2	4	5	A, B, C, D	(A–B, 1), (B–C, 2), (C–D, 3)

3. Results

The following table presents the MST results for each algorithm on each graph:

Graph ID	Algorithm	MST Cost	Operations	Time (ms)	MST Edges
1	Prim	16	42	1.52	B–C(2), A–C(3), B–D(5), D–E(6)
1	Kruskal	16	37	1.28	B–C(2), A–C(3), B–D(5), D–E(6)

2	Prim	6	29	0.87	A-B(1), B-C(2), C-D(3)
2	Kruskal	6	31	0.92	A-B(1), B-C(2), C-D(3)

4. Comparative Analysis

A comparison of the two algorithms is shown below:

Aspect	Prim's Algorithm	Kruskal's Algorithm	Comments
Ease of Implementation	Easy	Moderate	Kruskal requires Union-Find
Performance on Dense Graphs	Better	Worse	Prim is optimal for dense graphs
Performance on Sparse Graphs	Worse	Better	Kruskal performs better when fewer edges
Data Structures Used	Min-Heap	Union-Find	Different internal logic
Time Complexity	$O(E \log V)$	$O(E \log E)$	Similar asymptotic efficiency
Execution Time (from JSON)	1.52 ms	1.28 ms	Kruskal slightly faster

5. Conclusion

Both Prim's and Kruskal's algorithms produced the same MST total cost, confirming correctness. Kruskal's algorithm demonstrated slightly better performance in execution time for the given graphs. Prim's algorithm, however, is more efficient for dense graphs due to its adjacency-based approach, while Kruskal's is more suitable for sparse graphs and is conceptually simpler when working with edge lists.

6. References

1. https://en.wikipedia.org/wiki/Minimum_spanning_tree
2. [Difference between Prim's and Kruskal's algorithm for MST - GeeksforGeeks](#)